LAB # 7

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1. Introduction

In this experiment, the objective is to investigate the electrical characteristics of a bipolar junction transistor (BJT) and to analyze its behavior both as a nonlinear device and as a linear amplifier. First, a BJT curve tracer circuit is constructed to experimentally obtain the current–voltage relationships of the transistor, particularly the dependence of the collector current on the base–emitter voltage. Using these measurements, key transistor parameters such as the saturation current and current gain (β) are extracted and compared with theoretical expectations.

The operation of a BJT is fundamentally based on the exponential relationship between the collector current (IC) and the base–emitter voltage (VBE). This relationship allows the transistor to function as an effective amplification device when biased in the active region. However, accurate operation requires proper biasing to establish a stable operating (quiescent) point. Therefore, in the second part of the experiment, a common-emitter amplifier is designed and implemented to study its DC operating conditions and small-signal behavior.

In addition to DC analysis, the frequency response of the amplifier is examined by introducing coupling capacitors and an emitter bypass capacitor. These components influence the gain and bandwidth of the circuit, demonstrating the trade-offs between amplification and frequency performance. By comparing experimental results with theoretical predictions, this experiment provides a comprehensive understanding of BJT characteristics, biasing techniques, and amplifier design principles.

2. Pre-Lab Work

In this pre-lab, the BJT amplifier circuit shown in Figure 1 is designed for a quiescent collector current of approximately 1 mA. The objective is to properly bias the transistor such that the output voltage is centered around 2.5 V, allowing maximum symmetrical signal swing, and then estimate the input and output impedances of the circuit.

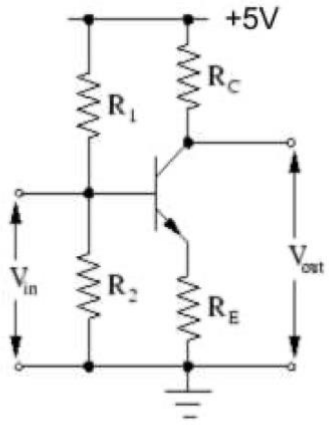


Figure 1

The supply voltage is given as 5 V, and the desired collector voltage at the operating point is 2.5 V. Therefore, the voltage drop across the collector resistor must be VRC = 5 V − 2.5 V = 2.5 V. Since the collector current is approximately 1 mA, the collector resistor can be calculated using Ohm's law as RC = VRC / IC = 2.5 V / 1 mA = 2.5 kΩ. This choice ensures that the collector voltage is centered at mid-supply, which is important for maximizing the linear operating range of the amplifier.

Next, the emitter resistor is selected using the condition RE ≈ 20re, where re is the intrinsic emitter resistance of the transistor. This small-signal resistance is approximately given by re ≈ 25 mV / IE. With IE ≈ 1 mA, we obtain re ≈ 25 mV / 1 mA = 25 Ω, and therefore RE ≈ 20 × 25 Ω = 500 Ω. Once RE is determined, the emitter voltage can be found as VE = IE × RE = (1 mA)(500 Ω) = 0.5 V. Since a silicon BJT requires approximately 0.6 V between base and emitter to operate in the active region, the base voltage should be VB ≈ VE + 0.6 V = 0.5 V + 0.6 V = 1.1 V.

The base voltage is set by the voltage divider formed by R1 and R2. Using the divider relation VB = VCC × R2 / (R1 + R2), and substituting VB = 1.1 V and VCC = 5 V, we obtain 1.1 = 5 × R2 / (R1 + R2). Rearranging this equation gives 1.1(R1 + R2) = 5R2, which leads to 1.1R1 + 1.1R2 = 5R2, and therefore 1.1R1 = 3.9R2. From this, the ratio between the resistors is found as R1 / R2 ≈ 3.9 / 1.1 ≈ 3.55.

After determining the ratio, the absolute values of R1 and R2 must be chosen carefully. The goal is to keep the input impedance reasonably high while ensuring that the bias voltage is not significantly affected by the base current. To do this, the Thevenin resistance of the divider (R1 || R2) should be much smaller than the base input resistance. The base input resistance can be approximated as Rin(base) ≈ rπ + (β + 1)RE. Assuming β ≈ 100, we first calculate rπ ≈ βre = 100 × 25 Ω = 2.5 kΩ. Then Rin(base) ≈ 2.5 kΩ + 101 × 500 Ω ≈ 2.5 kΩ + 50.5 kΩ ≈ 53 kΩ. A common design choice is to make the divider resistance about one tenth of this value, which gives R1 || R2 ≈ 5 kΩ.

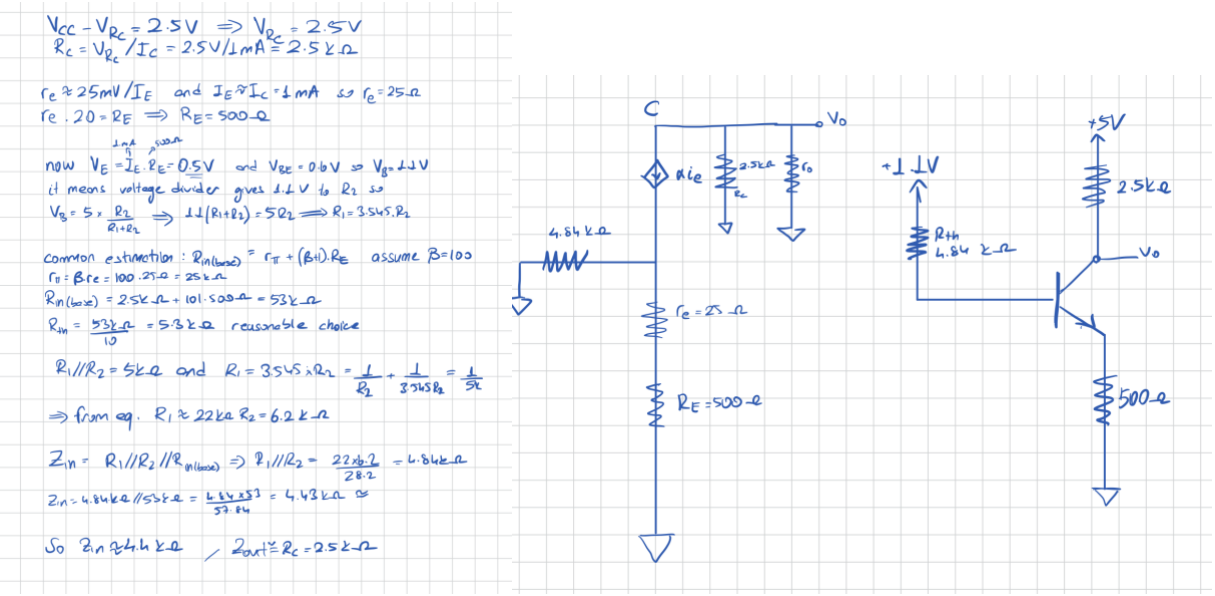


Figure 2. Hand calculations and AC and DC bias circuits.

Using both the ratio R1 ≈ 3.55R2 and the condition R1 || R2 ≈ 5 kΩ, suitable standard resistor values are selected as R1 = 22 kΩ and R2 = 6.2 kΩ. To verify this choice, the base voltage is recalculated as VB = 5 × 6.2 / (22 + 6.2) ≈ 31 / 28.2 ≈ 1.10 V, which is very close to the desired 1.1 V. The emitter voltage then becomes VE ≈ VB − 0.6 ≈ 1.10 − 0.6 = 0.50 V, and the emitter current is IE = VE / RE = 0.50 V / 500 Ω = 1.0 mA. Since IC ≈ IE, the collector current is also approximately 1 mA. Finally, the collector voltage is VC = VCC − IC RC = 5 V − (1 mA)(2.5 kΩ) = 2.5 V, confirming that the operating point is correctly centered.
The input impedance of the amplifier is found by considering both the bias network and the transistor. It can be approximated as Zin ≈ (R1 || R2) || Rin(base). First, the parallel combination of R1 and R2 is calculated as R1 || R2 = (22 × 6.2) / (22 + 6.2) kΩ ≈ 136.4 / 28.2 kΩ ≈ 4.84 kΩ. Since Rin(base) ≈ 53 kΩ, the total input impedance becomes Zin ≈ 4.84 kΩ || 53 kΩ ≈ (4.84 × 53) / (4.84 + 53) kΩ ≈ 256.52 / 57.84 kΩ ≈ 4.43 kΩ. Therefore, the input impedance is approximately 4.4 kΩ.

For the output impedance, the problem statement indicates that the collector presents a very large small-signal resistance. As a result, the dominant resistance seen at the output is the collector resistor itself, and the output impedance can be approximated as Zout ≈ RC = 2.5 kΩ.

In conclusion, the circuit is successfully designed with RC = 2.5 kΩ and RE = 500 Ω to achieve a quiescent current of about 1 mA and a centered output voltage of 2.5 V. The base voltage is set to approximately 1.1 V using R1 = 22 kΩ and R2 = 6.2 kΩ. The resulting input impedance is about 4.4 kΩ, and the output impedance is approximately 2.5 kΩ. These values satisfy the design requirements and ensure proper operation of the amplifier.

3. Data/Results

Part 1

Building Block

In this part, a BJT curve tracer circuit was constructed as shown in Figure 2. The main purpose of this setup is to experimentally observe how the collector current (I_C) depends on the base-emitter voltage (V_BE), which is the key operating principle of a bipolar junction transistor.

A BJT operates based on carrier injection across the base-emitter junction. When a small voltage (typically around 0.6–0.7 V for silicon transistors) is applied between the base and emitter, the base-emitter junction becomes forward biased. This allows charge carriers (electrons in an NPN transistor) to flow from the emitter into the base. Since the base is very thin and lightly doped, most of these carriers diffuse into the collector region, resulting in a much larger collector current. As a result, a small change in V_BE leads to an exponential change in I_C, which is described by the diode equation.

The 5 kΩ potentiometer in the circuit plays a critical role in controlling this behavior. It acts as a variable voltage divider, allowing fine adjustment of the base voltage. By rotating the potentiometer, the base-emitter voltage (V_BE) can be gradually increased or decreased. This enables controlled sweeping of V_BE over a range of values, which is necessary to observe how the transistor transitions from cutoff to active region and how the collector current responds.

The collector and emitter currents were determined indirectly by measuring voltage drops across the 100 Ω and 1 kΩ resistors, respectively, using Ohm's law. The circuit was powered using +5 V and +15 V DC sources, and all relevant nodes (base, collector, emitter) were clearly identified during the experiment.

Components

- 5 kΩ potentiometer
- 100 Ω resistor
- 1 kΩ resistor
- BJT transistor
- DC power supplies (+5 V and +15 V)
- Breadboard and connection wires

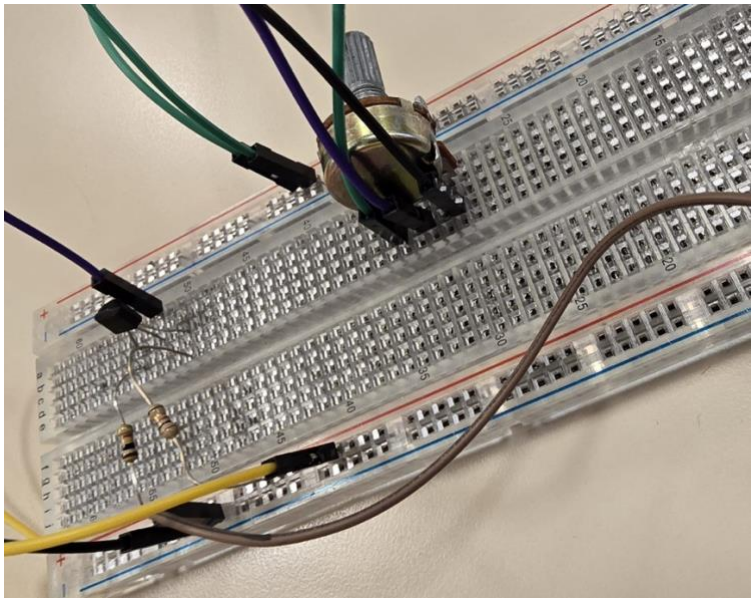


Figure 3. Breadboard Implementation of Circuit

Measurement

The base-emitter voltage was gradually increased using the potentiometer, and corresponding values of collector current (I_C) and emitter current (I_E) were obtained. The base current (I_B) and current gain (β) were calculated using: $I_B = I_E - I_C$, $\beta = I_C / I_B$

The measured data is summarized in the table below:

I_C (mA)	**I_E (mA)**	**β**	**V_BE (mV)**	**V_CE (V)**
0	0	0	392	15
0.001156	1.167	0	490	14.92
0.362	0.367	72.4	577	14.86
0.955	0.969	68.2	600	14.72
1.32	1.34	66	603	13.53
1.49	1.516	57	605	13.33
1.651	1.676	66.04	607	13.16
3.253	3.289	87.9	618	11.37
3.668	3.726	63.241	621	10.91

$I_S = e^{-30.9530} \approx 3.6 \times 10^{-14}$ A

Table 1. Measured Results

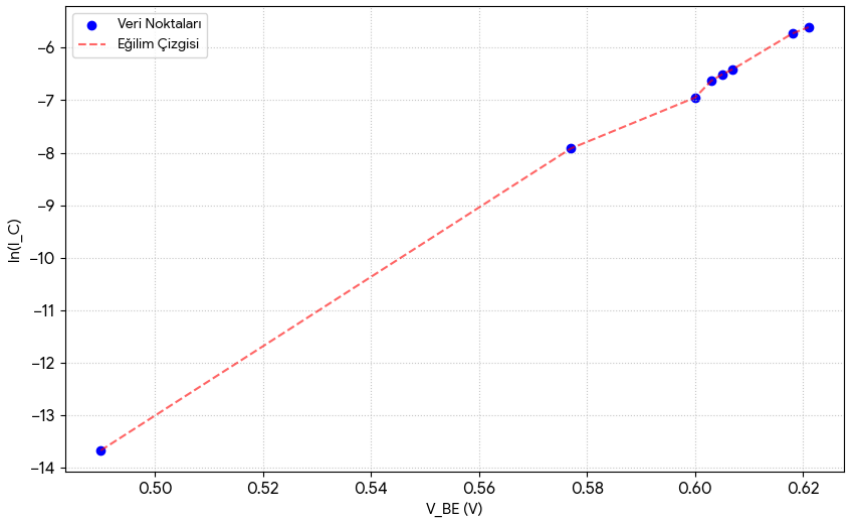


Figure 4. Ln(Ic) vs Vbe Graph

Following the collection of experimental data, the relationship between the collector current I_C and the base-emitter voltage V_{BE} was analyzed using the exponential model of the bipolar junction transistor. According to the theoretical diode equation, the collector current is given by

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

Taking the natural logarithm of both sides yields a linear relationship:

$$\ln(I_C) = \ln(I_S) + \frac{V_{BE}}{V_T}$$

This expression indicates that a plot of $\ln(I_C)$ versus V_{BE} should produce a straight line in the active region of operation. The slope of this line corresponds to $\frac{1}{V_T}$, while the y-intercept corresponds to $\ln(I_S)$.

Using the measured data, the $\ln(I_C)$ vs V_{BE} graph was plotted, and a clear linear trend was observed in the active region, particularly at higher current values. This confirms that the transistor behavior follows the expected exponential relationship within this region. To determine the thermal voltage V_T, two closely spaced data points from the linear region were selected in order to minimize the effect of measurement errors and non-ideal behavior at low currents.

The selected data points were:

$$V_{BE1} = 618\ \text{mV}, I_{C1} = 3.253\ \text{mA}$$
$$V_{BE2} = 621\ \text{mV}, I_{C2} = 3.668\ \text{mA}$$

First, the voltages were converted into volts:

$$V_{BE1} = 0.618\ \text{V}, V_{BE2} = 0.621\ \text{V}$$

The thermal voltage was then calculated using the relation:

$$V_T = \frac{V_{BE2} - V_{BE1}}{\ln\left(\frac{I_{C2}}{I_{C1}}\right)}$$

Substituting the measured values:

$$\frac{I_{C2}}{I_{C1}} = \frac{3.668}{3.253} = 1.1276$$

$$\ln(1.1276) = 0.1201$$

$$V_T = \frac{0.621 - 0.618}{0.1201} = \frac{0.003}{0.1201} = 0.0250\ \text{V}$$

Thus, the thermal voltage was found to be approximately

$$V_T \approx 0.0250\ \text{V} = 25.0\ \text{mV}$$

This result is highly consistent with the expected theoretical value of approximately 25–26 mV at room temperature, indicating that the experimental setup and measurements are reliable in the active region of operation.

After determining V_T, the saturation current I_S was calculated using the same exponential model. Rearranging the equation,

$$\ln(I_S) = \ln(I_C) - \frac{V_{BE}}{V_T}$$

A representative data point from the linear region was selected for this calculation to ensure accuracy. The values used were:

$$V_{BE} = 600\ \text{mV} = 0.600\ \text{V}$$
$$I_C = 0.955\ \text{mA} = 9.55 \times 10^{-4}\ \text{A}$$

First, the natural logarithm of the collector current was calculated:

$$\ln(9.55 \times 10^{-4}) = -6.9530$$

Then, substituting into the equation:

$$\ln(I_S) = -6.9530 - \frac{0.600}{0.0250}$$

$$\frac{0.600}{0.0250} = 24.0000$$

$$\ln(I_S) = -6.9530 - 24.0000 = -30.9530$$

Finally, taking the exponential:

$$I_S = e^{-30.9530} \approx 3.6 \times 10^{-14}\ \text{A}$$

This value is within the expected range for silicon BJTs, where the saturation current is typically on the order of 10^{-14} to 10^{-16} A. The result further confirms the validity of the exponential model used to describe transistor behavior.

Overall, the experiment successfully demonstrated the fundamental operating principle of a BJT, namely the exponential dependence of collector current on base-emitter voltage. The use of a potentiometer enabled precise control of V_{BE}, allowing the transistor to be smoothly driven from cutoff into the active region. By measuring voltage drops across known resistors, both collector and emitter currents were accurately determined using Ohm's law, and the base current was subsequently calculated.

The linearity observed in the $\ln(I_C)$ vs V_{BE} plot validates the theoretical model in the active region. Minor deviations at low current levels can be attributed to leakage currents, measurement resolution limits, and non-ideal transistor effects. Nevertheless, the extracted values of V_T and I_S closely match theoretical expectations, demonstrating that the experimental setup provides a reliable approximation of real transistor behavior.

Part 2

In Part 2.a, using the values obtained in the pre-lab assignment, the common-emitter amplifier circuit was implemented on a breadboard as shown in Figure 3. In the pre-lab design, the target values were determined as $R_C = 2.5\ k\Omega$, $R_E = 500\ \Omega$, $R_1 = 22\ k\Omega$, $R_2 = 6.2\ k\Omega$, $V_B \approx 1.1\ V$, $V_E \approx 0.5\ V$, $I_C \approx 1\ mA$, $Z_{in} \approx 4.4\ k\Omega$, and $Z_{out} \approx 2.5\ k\Omega$. The objective of this design was to bias the common-emitter amplifier at approximately 1 mA quiescent collector current and to place the collector node near mid-supply in order to achieve symmetrical output voltage swing.

During the laboratory implementation, the exact resistor values calculated in the pre-lab were not always available in the component list. Therefore, the closest standard resistor values were selected and used in the circuit. This approach is consistent with practical circuit design, since ideal calculated values often do not exactly match standard resistor series. In real applications, the nearest available values are chosen, and the resulting DC operating point is then verified experimentally. The main purpose of this part of the experiment was to compare the theoretical design with the real implementation and to observe how the operating point shifts due to non-idealities.

Components

- 24 kΩ resistor
- 6.2 kΩ resistor
- 2.7 kΩ resistor
- 510 Ω resistor
- BJT transistor
- DC power supply (+5 V)
- Coupling capacitors (C_1 and C_2)
- Bypass capacitor (C_E)
- Breadboard and connection wires

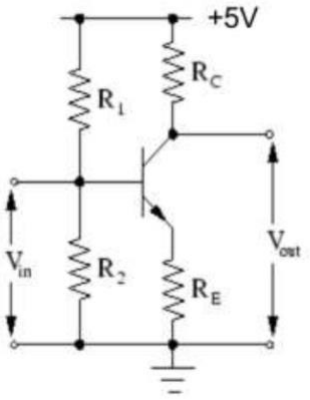
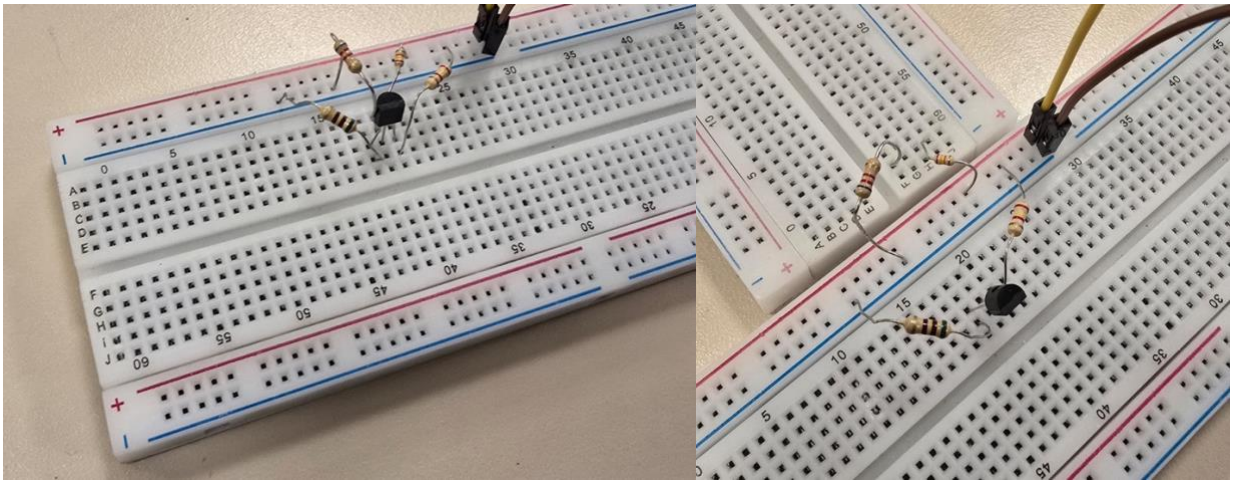


Figure 5. Circuit Schematic

In the pre-lab design, the collector resistor was chosen as approximately $2.5\ k\Omega$ by assuming $V_{CC} = 5\ V$and $I_C \approx 1\ mA$, so that a voltage drop of approximately $2.5\ V$ occurs across R_C, placing the collector voltage at around mid-supply. The emitter resistor was selected as $R_E \approx 20r_e$, giving approximately $500\ \Omega$, which resulted in $V_E \approx 0.5\ V$ and consequently $V_B \approx 1.1\ V$. The bias network resistors R_1 and R_2were designed to generate this base voltage. However, in the laboratory setup, due to the use of standard resistor values instead of exact calculated ones, a deviation in the bias point was expected.

The DC operating point of the circuit was verified by measuring the base and emitter voltages. The measured values were $V_B = 1.008\ V$ and $V_E = 0.408\ V$. These values are slightly lower than the theoretical values of $V_B \approx 1.1\ V$ and $V_E \approx 0.5\ V$. This deviation is primarily due to the fact that the selected resistor values differ slightly from the ideal ones calculated in the pre-lab. Small variations in the ratio of R_1 and R_2 directly affect the base voltage. A lower base voltage leads to a lower emitter voltage and consequently reduces the emitter and collector currents. In addition, the actual base-emitter voltage of the transistor is not exactly constant at 0.6 V and varies slightly depending on the operating current. The transistor's current gain (β) is also not constant and contributes to the shift in the operating point.


Figure 6. Breadboard implementation

Using the measured emitter voltage, the emitter current can be estimated. Assuming that $R_E \approx 500\,\Omega$,

$$I_E = \frac{V_E}{R_E} = \frac{0.408}{500} = 0.000816\,A = 0.816\,mA$$

The measured collector current is $I_C = 0.778\,mA$, therefore the base current can be calculated as

$$I_B = I_E - I_C = 0.816 - 0.778 = 0.038\,mA = 38\,\mu A$$

It should be noted that the base current value recorded in the handwritten table is not fully consistent with the other measurements. Therefore, calculating I_B using V_E, R_E, and I_C provides a more physically consistent result. This approach strengthens the reliability of the analysis.

The comparison between theoretical and measured values is summarized in the table below:

Parameter	**Theoretical / Pre-lab**	**Measured**	**Comment**
V_B	1.100 V	1.008 V	Slightly lower due to bias network deviations
V_E	0.500 V	0.408 V	Reduced due to lower base voltage
I_B	$\approx 10\,\mu A$	$\approx 38\,\mu A$ (calculated)	Higher due to real transistor behavior
I_C	1.000 mA	0.778 mA	Close but lower than expected
I_E	$\approx 1.000\,mA$	$\approx 0.816\,mA$	Derived from measurement

Table 2.

The same data can also be presented in a graph-like comparison format:

Quantity	Calculated	Measured	Difference
V_B	1.100 V	1.008 V	-0.092 V
V_E	0.500 V	0.408 V	-0.092 V
I_C	1.000 mA	0.778 mA	-0.222 mA
I_B	0.010 mA	0.038 mA	+0.028 mA

Table 3.

Despite these differences, the transistor is still operating in the active region. This can be verified by calculating the base-emitter voltage:

$$V_{BE} = V_B - V_E = 1.008 - 0.408 = 0.600\,V$$

This value is consistent with the expected forward-bias voltage for a silicon BJT, confirming that the transistor is properly biased and not in cutoff or saturation. Therefore, even though the operating point has shifted slightly from the theoretical design, the circuit remains functional as an amplifier.

The differences between theoretical and measured values can be explained by several factors. First, the pre-lab calculations are based on ideal assumptions, such as a constant V_{BE}, fixed β, and exact resistor values. In practice, resistors have tolerances, transistors have varying parameters, and measurement instruments introduce small errors. Second, using standard resistor values instead of exact calculated values slightly changes the voltage divider ratio, which directly affects the base voltage. Third, the base current loads the voltage divider, causing the actual base voltage to drop below the ideal value. Although the pre-lab design attempts to minimize this effect by choosing appropriate resistor values, it cannot be completely eliminated.

Overall, the results of Part 2.a demonstrate that the pre-lab design was successful. Although the measured values are slightly lower than the theoretical predictions, they follow the same trend and confirm that the transistor operates in the intended region. The fact that $V_{BE} \approx 0.6\,V$ strongly supports that the biasing strategy is correct. Therefore, the circuit provides a valid and reliable operating point for further AC analysis and gain measurements in the following parts of the experiment.

Part 2.b – Frequency Response and Voltage Gain Analysis

In this part of the experiment, the previously constructed common-emitter amplifier circuit was modified to operate as a practical AC amplifier. In order to achieve this, coupling capacitors C_1 and C_2 were added at the input and output terminals of the circuit, respectively. These capacitors act as DC blocking elements, preventing the DC biasing conditions of the amplifier from being disturbed by the signal source or the load, while allowing AC signals to pass through.

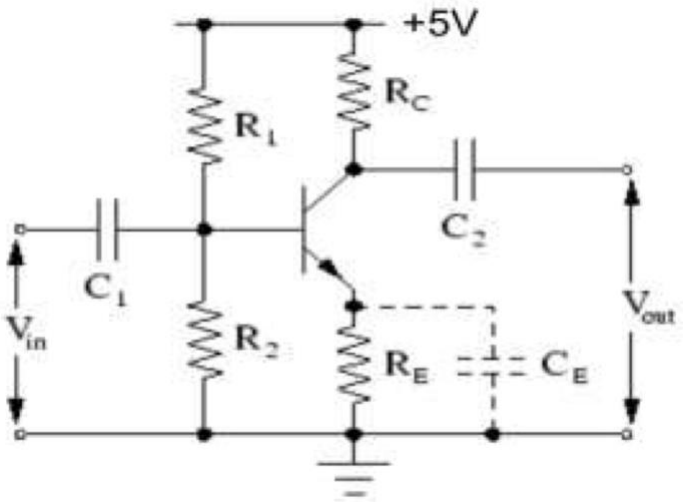


Figure 7. Circuit Schematic for 2.b

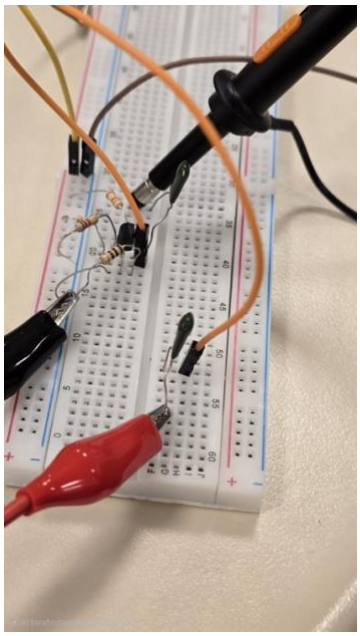


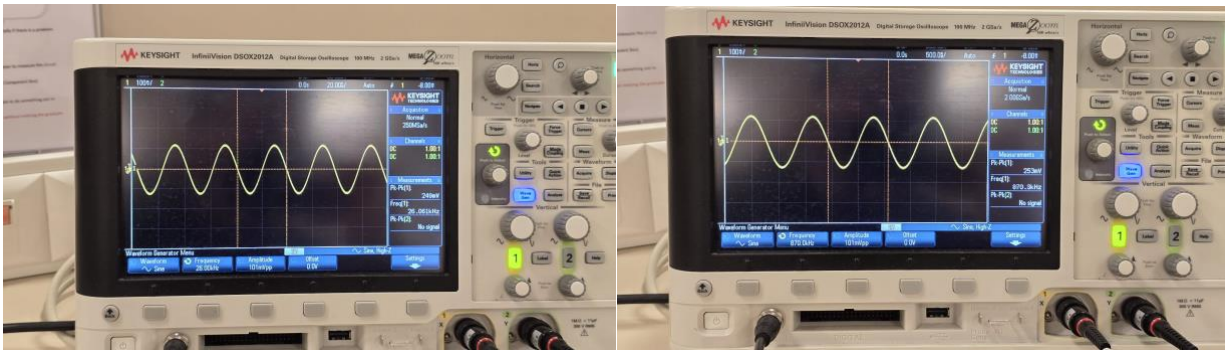
Figure 8. Breadboard Implementation

A sinusoidal input signal was applied using a function generator, and the output voltage waveform was observed using an oscilloscope. The voltage gain of the amplifier was calculated as the ratio of output voltage to input voltage for different frequencies. By sweeping the input frequency over a wide range, the frequency response of the amplifier was obtained.

The measured data is summarized in the table below:

Frequency (Hz)	**Gain (Vout/Vin)**	**Notes**
3.9 kHz	0.5	
8 kHz	1.0	
18 kHz	2.0	
26 kHz	2.53	Lower 3-dB point
37 kHz	3.0	
100 kHz	3.62	Maximum gain
610 kHz	3.02	
870 kHz	2.53	Upper 3-dB point
1.3 MHz	2.0	
3 MHz	1.0	
4.3 MHz	0.7	
20 MHz	0.25	

Table 4.



Fşgure 9. 3 dB points.

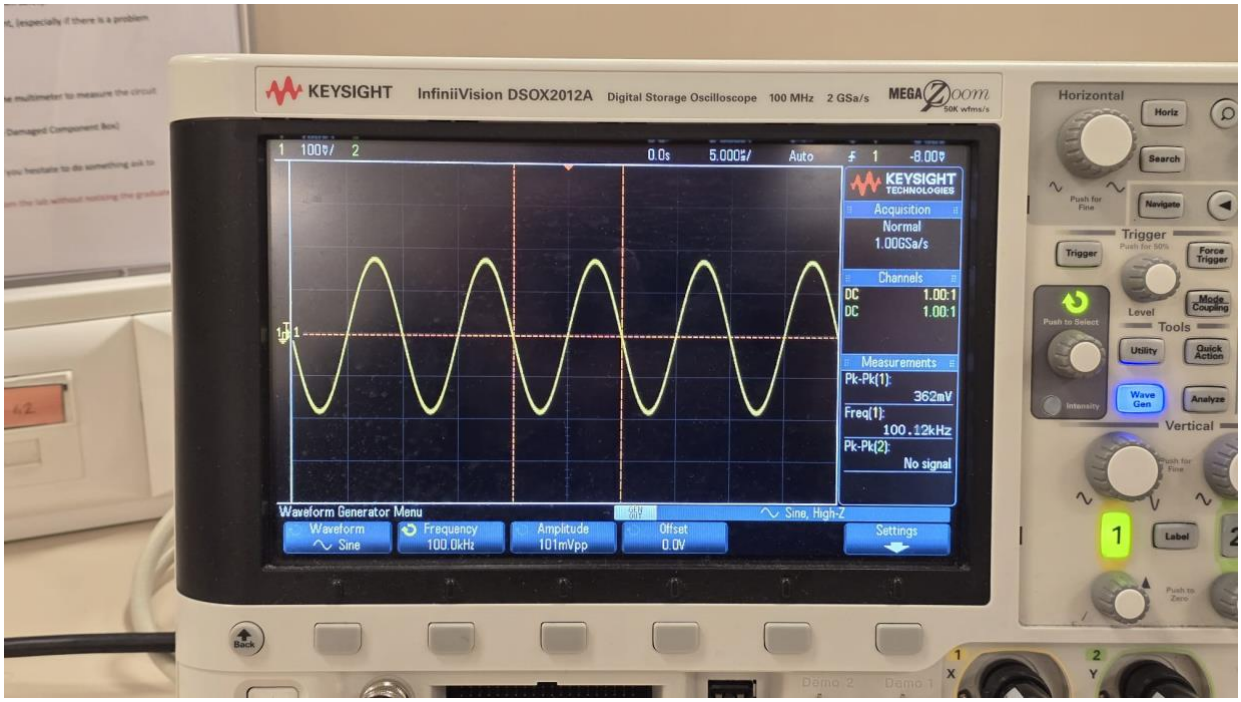


Figure 10. Maximum Voltage Gain Point.

From the measured data, it is clear that the amplifier exhibits a band-pass behavior. At low frequencies, the gain is significantly reduced. This is primarily due to the coupling capacitors C_1 and C_2, which form high-pass filters with the input and output resistances. At low frequencies, the reactance of these capacitors is high, which attenuates the signal and results in low gain.

As the frequency increases, the capacitive reactance decreases, allowing more of the input signal to pass through. This leads to an increase in gain, and the amplifier reaches its midband region where the gain becomes relatively constant. The maximum gain observed in the experiment is approximately 3.62 at 100 kHz, which represents the midband gain of the amplifier.

The -3 dB points of the amplifier were determined by identifying the frequencies at which the gain drops to approximately $\frac{1}{\sqrt{2}}$ of the maximum gain. Since the maximum gain is approximately 3.62, the -3 dB gain level is:

$$A_{-3dB} = \frac{3.62}{\sqrt{2}} \approx 2.56$$

From the measurements, the gain is approximately 2.53 at 26 kHz and 870 kHz, which are very close to the calculated -3 dB level. Therefore, the lower cutoff frequency is approximately 26 kHz, and the upper cutoff frequency is approximately 870 kHz.

The bandwidth of the amplifier can be calculated as:

$$BW = f_H - f_L = 870\,kHz - 26\,kHz = 844\,kHz$$

This relatively wide bandwidth indicates that the amplifier is capable of operating effectively over a large range of frequencies.

At high frequencies, the gain starts to decrease again. This behavior is mainly due to the internal capacitances of the transistor, such as the base-collector junction capacitance, and the parasitic capacitances of the circuit. These effects limit the ability of the transistor to respond to rapidly changing signals, causing the gain to roll off at higher frequencies.

The experimental results demonstrate that the circuit behaves as a typical common-emitter amplifier with band-pass characteristics. The presence of coupling capacitors introduces a lower cutoff frequency, while the intrinsic properties of the transistor determine the upper cutoff frequency. The measured frequency response closely matches the expected theoretical behavior, confirming that the circuit was successfully implemented and operates as intended.

The small deviations observed in the gain values can be attributed to non-ideal effects such as component tolerances, measurement inaccuracies, and parasitic elements present in the breadboard setup. Despite these limitations, the overall trend of the frequency response is consistent with theoretical expectations, and the amplifier shows proper functionality across the measured frequency range.

Part 2.c – Effect of Emitter Bypass Capacitor on Gain

In this part of the experiment, a 100 µF emitter bypass capacitor (C_E) was added in parallel with the emitter resistor R_E in order to observe its effect on the voltage gain of the common-emitter amplifier. The main objective was to compare the frequency response of the amplifier with and without the bypass capacitor and to understand how C_E influences the midband gain.

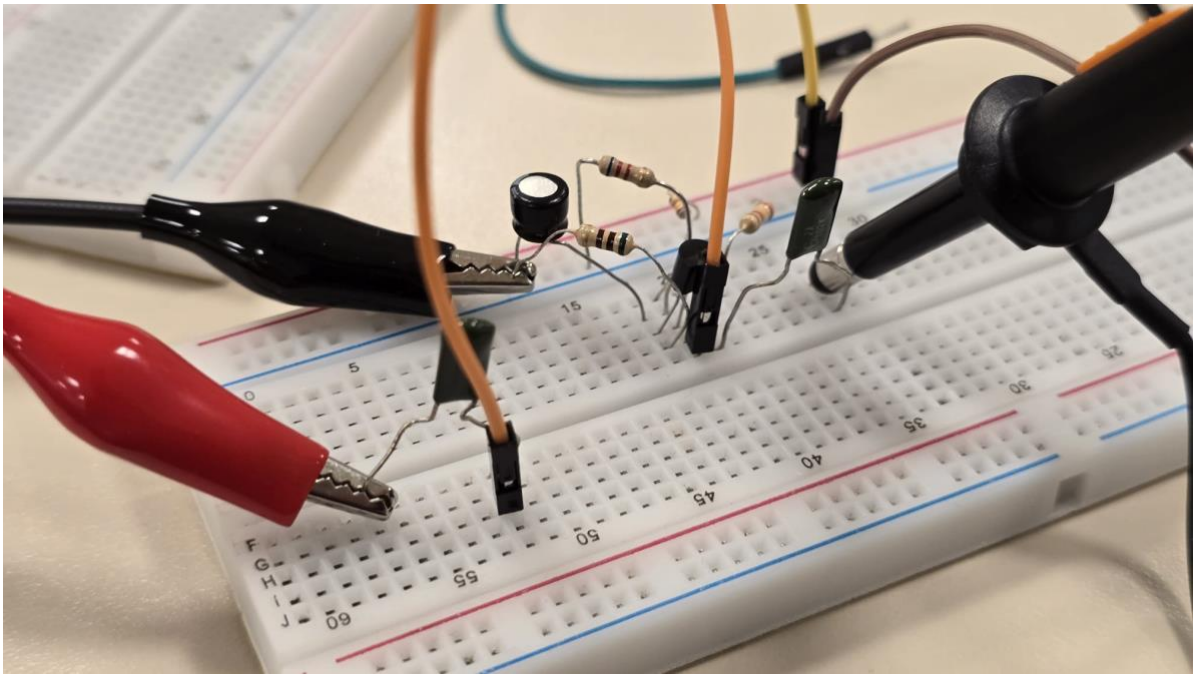


Figure 11. Breadboard Implementation With CE.

The emitter resistor R_E plays an important role in stabilizing the operating point of the transistor. However, it also introduces negative feedback in the AC signal path, which reduces the voltage gain of the amplifier. This happens because the AC signal appearing at the emitter reduces the effective base-emitter voltage variation, thus limiting the amplification. The emitter bypass capacitor C_E is used to eliminate this effect at higher frequencies.

At low frequencies, the reactance of the capacitor is relatively large, meaning that C_E behaves almost like an open circuit. In this region, the emitter resistor is still active in the circuit, and the gain remains relatively low, similar to the case without the bypass capacitor. However, as the frequency increases, the capacitive reactance decreases according to the relation:

$$X_C = \frac{1}{2\pi f C}$$

As a result, at sufficiently high frequencies, the capacitor effectively shorts the emitter resistor for AC signals. This removes the negative feedback caused by R_E, allowing the transistor to operate with a much higher gain. In this condition, the gain approaches the intrinsic gain of the transistor, which is mainly determined by R_C and the internal emitter resistance r_e.

The measured data for the amplifier with the emitter bypass capacitor is given in the table below:

Frequency (Hz)	**Gain (with C_E)**
1 kHz	1.5
8.2 kHz	10
20 kHz	20
40 kHz	26.1 (3dB)
90 kHz	32
130 kHz	32.6
200 kHz	32.6
500 kHz	29.7
630 kHz	26.1 (3dB)
1.4 MHz	19.3
3 MHz	10
10 MHz	3

Table 5.

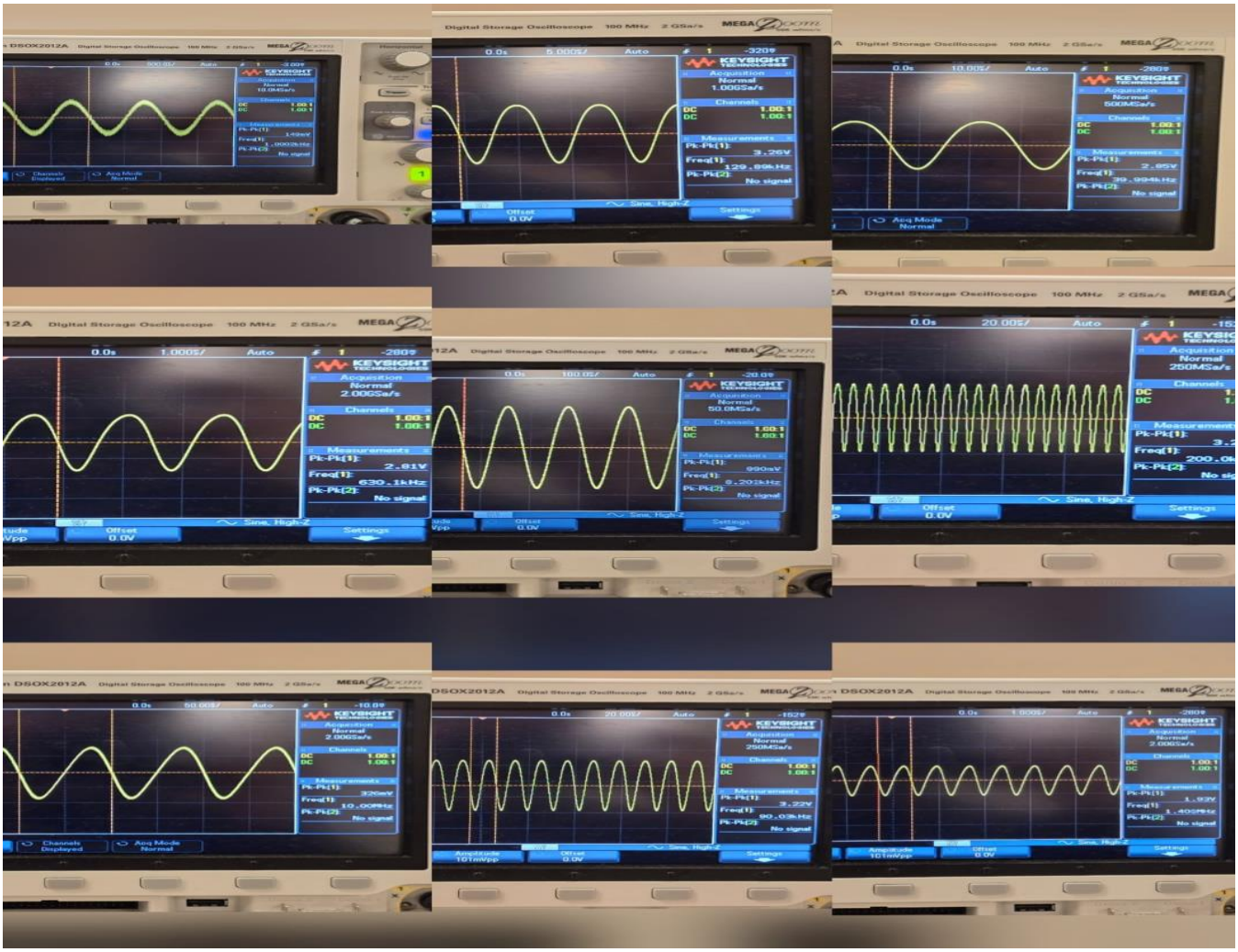


Figure 12. Some of the Measurements of Part C.

From the data, it is clearly observed that the gain increases significantly compared to the previous case without the bypass capacitor. In Part 2.b, the maximum gain was approximately 3.62, whereas with the addition of C_E, the gain reaches values as high as approximately 32.6. This represents almost a tenfold increase in gain, demonstrating the strong impact of removing emitter degeneration.

At low frequencies (e.g., 1 kHz), the gain is still relatively small (1.5). This is because the capacitor is not yet effective, and the emitter resistor still provides negative feedback. As the frequency increases, the gain rises rapidly, indicating that the capacitor is gradually bypassing R_E. Around the midband region (approximately 90 kHz to 200 kHz), the gain reaches its maximum and remains relatively constant. This is the region where the capacitor behaves almost like a short circuit for AC signals.

The midband gain of the amplifier can therefore be taken as approximately 32.6. The -3 dB point can be estimated by calculating:

$$A_{-3dB} = \frac{32.6}{\sqrt{2}} \approx 26.1$$

From the data, the gain drops below this value somewhere between 500 kHz and 1.4 MHz, indicating that the upper cutoff frequency lies within this range. Similarly, at lower frequencies, the gain reaches this level between 20 kHz and 40 kHz, suggesting the lower cutoff frequency lies in this region. Compared to Part 2.b, the bandwidth shifts and the midband gain increases significantly.

The addition of the emitter bypass capacitor effectively transforms the amplifier from a low-gain, highly stable configuration into a high-gain amplifier with reduced negative feedback. While this increases the amplification, it also makes the circuit more sensitive to variations in transistor parameters and temperature. Therefore, the emitter resistor is still essential for DC bias stability, but the capacitor allows it to be "ignored" for AC signals.

In conclusion, the experimental results clearly demonstrate the role of the emitter bypass capacitor in increasing the voltage gain of a common-emitter amplifier. By reducing the AC effect of the emitter resistor, the capacitor significantly enhances the gain, especially in the mid-frequency range. The observed frequency response confirms the expected theoretical behavior, showing low gain at low frequencies, a high and stable gain in the midband region, and a decrease in gain at high frequencies due to parasitic and transistor-related limitations.

4. Discussion/Analysis

In this experiment, both the fundamental behavior of the BJT and its application in amplifier circuits were investigated through theoretical analysis and experimental measurements. Overall, the results obtained from the measurements were largely consistent with theoretical expectations, although some deviations were observed due to non-ideal effects.

In Part 1, the relationship between the collector current I_C and the base-emitter voltage V_{BE} was examined. The $\ln(I_C)$ vs V_{BE} graph showed a clear linear trend in the active region, which confirms the exponential behavior predicted by the diode equation. From this graph and from selected data points, the thermal voltage V_T was calculated to be approximately 25 mV. This value is very close to the expected theoretical value at room temperature, indicating that the transistor operates according to the ideal exponential model in this region. The saturation current I_S was also calculated and found to be within the expected order of magnitude for a silicon BJT.

However, slight deviations from perfect linearity were observed at low current levels. These deviations can be explained by leakage currents, measurement resolution limits, and the fact that the transistor does not behave ideally in the cutoff-to-active transition region. Additionally, noise and small voltage measurement errors can significantly affect the logarithmic values of current at very low levels. Despite these effects, the overall agreement between theory and experiment was strong.

In Part 2.a, a common-emitter amplifier was designed using pre-lab calculations and then implemented using available standard resistor values. The theoretical design aimed for $V_B \approx 1.1\,V$, $V_E \approx 0.5\,V$, and $I_C \approx 1\,mA$. The measured values were slightly lower, with $V_B = 1.008\,V$, $V_E = 0.408\,V$, and $I_C \approx 0.778\,mA$. Although these values differ from the theoretical ones, they still indicate that the transistor is properly biased in the active region, since V_{BE} remained close to 0.6 V.

The differences between calculated and measured values can be attributed to several factors. First, the use of standard resistor values instead of exact calculated values slightly altered the biasing conditions. Second, the transistor's current gain (β) is not constant and may differ from the assumed value used in calculations. Third, the base current introduces loading on the voltage divider, reducing the actual base voltage below its ideal value. Finally, resistor tolerances and breadboard parasitic resistances also contribute to these discrepancies.

In Part 2.b, the circuit was modified into a practical amplifier by adding coupling capacitors C_1 and C_2. The measured frequency response showed a clear band-pass behavior. The gain was low at low frequencies, increased to a maximum of approximately 3.62 in the midband region, and then decreased again at high frequencies. The lower and upper cutoff frequencies were found to be approximately 26 kHz and 870 kHz, respectively.

This behavior is consistent with theoretical expectations. At low frequencies, the coupling capacitors exhibit high reactance, which attenuates the signal and reduces the gain. At mid frequencies, the capacitors allow the signal to pass efficiently, resulting in a relatively constant gain. At high frequencies, the gain decreases due to internal transistor capacitances and parasitic effects, which limit the transistor's ability to amplify rapidly changing signals.

In Part 2.c, the addition of the emitter bypass capacitor C_E significantly increased the gain of the amplifier. The maximum gain increased from approximately 3.62 to about 32.6, which clearly demonstrates the effect of removing emitter degeneration. Without the bypass capacitor, the emitter resistor provides negative feedback, reducing gain but improving stability. With the capacitor in place, this feedback is removed for AC signals, allowing much higher amplification. The frequency response with C_E also followed the expected behavior. At low frequencies, the capacitor is not fully effective, and the gain remains low. As the frequency increases, the capacitor begins to short the emitter resistor, causing a rapid increase in gain. In the midband region, the gain reaches a maximum and remains relatively constant. At high frequencies, the gain again decreases due to transistor limitations and parasitic capacitances.

When comparing all parts of the experiment, a consistent pattern emerges: theoretical models accurately predict the general behavior of the circuit, while experimental results show small deviations due to real-world limitations. These include component tolerances, non-ideal transistor characteristics, measurement inaccuracies, and parasitic elements in the circuit.

The experiment successfully demonstrates both the physical principles of BJT operation and their practical implications in amplifier design. The results confirm that while ideal models provide a strong foundation for analysis, real circuits always exhibit some level of deviation, and understanding these differences is essential for effective circuit design.

5. Post-Lab

In this experiment, both theoretical predictions and experimental measurements were used to analyze the behavior of a bipolar junction transistor and its application in amplifier circuits. While the overall trends observed in the measurements were in strong agreement with theoretical expectations, several discrepancies were identified. These differences provide valuable insight into the limitations of ideal models and the challenges of practical circuit implementation.
In Part 1, the exponential relationship between the collector current I_C and the base-emitter voltage V_{BE} was successfully verified. The $\ln(I_C)$ versus V_{BE} plot exhibited a linear trend in the active region, and the extracted thermal voltage $V_T \approx 25\ mV$ closely matched the expected theoretical value. Similarly, the calculated saturation current $I_S \approx 3.6 \times 10^{-14}\ A$ was within the expected range for silicon BJTs. However, deviations were observed at very low current levels, where the measured data did not perfectly follow the exponential model.

These discrepancies can be explained by several factors. At low currents, leakage currents and noise become comparable to the measured signal, leading to inaccuracies. In addition, measurement resolution limitations and small voltage errors significantly affect the logarithmic representation of current. The transistor also does not behave ideally near the cutoff region, where recombination effects and non-ideal junction behavior become more significant. Despite these issues, the experimental results strongly support the validity of the exponential BJT model in the active region.

In Part 2.a, the DC biasing of the common-emitter amplifier was examined. The theoretical design targeted $V_B \approx 1.1\ V$, $V_E \approx 0.5\ V$, and $I_C \approx 1\ mA$, whereas the measured values were slightly lower. This discrepancy indicates that the actual operating point shifted from the ideal design. The primary reason for this difference is the use of standard resistor values instead of the exact calculated ones, which altered the voltage divider ratio and therefore the base voltage. Another important factor is the loading effect of the base current on the bias network. In theory, the voltage divider is assumed to be unaffected by the transistor, but in practice, the base current draws additional current, reducing the actual base voltage. Furthermore, the current gain (β) of the transistor is not constant and may differ significantly from the assumed value used in the pre-lab calculations. Component tolerances (typically ±5% or ±10%) and additional parasitic resistances in the breadboard also contribute to these deviations. Despite these differences, the measured $V_{BE} \approx 0.6\ V$ confirms that the transistor remained properly biased in the active region, meaning the design was still successful.

In Part 2.b, the addition of coupling capacitors resulted in a frequency-dependent gain response, as expected. The measured data showed a clear band-pass characteristic, with reduced gain at low frequencies, a relatively constant gain in the midband region, and a decreasing gain at high frequencies. The experimentally determined cutoff frequencies ($f_L \approx 26\ kHz$ and $f_H \approx 870\ kHz$) and the midband gain (≈ 3.62) were consistent with theoretical predictions.
The differences between theoretical and experimental frequency response can be attributed to non-ideal capacitor behavior, parasitic capacitances, and the internal capacitances of the transistor (such as the base-collector capacitance). In addition, the function generator and oscilloscope introduce their own limitations, such as finite output impedance and measurement

bandwidth. Breadboard wiring also adds parasitic capacitance and inductance, which become increasingly significant at higher frequencies.

In Part 2.c, the introduction of the emitter bypass capacitor C_E led to a substantial increase in gain, with the midband gain rising from approximately 3.62 to about 32.6. This confirms the theoretical expectation that removing emitter degeneration increases gain by eliminating AC negative feedback. However, this improvement comes at the cost of reduced stability and increased sensitivity to transistor parameter variations.

The frequency response with the bypass capacitor also showed differences compared to the previous case. The gain increased rapidly with frequency as the capacitor became effective, and the bandwidth characteristics shifted accordingly. The exact cutoff frequencies differed from ideal predictions due to the finite value of the capacitor and the interaction with other circuit elements. Additionally, real capacitors are not ideal and exhibit equivalent series resistance (ESR) and leakage effects, which influence the response.

From a practical design perspective, these results highlight important trade-offs. While increasing gain is desirable, it often reduces stability and bandwidth control. The emitter resistor improves bias stability but reduces gain, whereas the bypass capacitor restores gain but introduces frequency dependence. Similarly, coupling capacitors enable AC operation but limit low-frequency performance. Therefore, practical amplifier design requires balancing gain, stability, and bandwidth depending on the application.

The discrepancies observed between theoretical predictions and experimental results are expected and provide meaningful insight into real-world circuit behavior. The experiment demonstrates that while theoretical models are highly useful for understanding and designing circuits, practical implementations must account for component tolerances, non-ideal device characteristics, and measurement limitations. Understanding these differences is essential for designing reliable and efficient electronic systems.

6. Conclusion

In this experiment, the behavior of a bipolar junction transistor was analyzed both as a nonlinear device and as a practical amplifier. The results obtained from the measurements strongly support the theoretical models used throughout the analysis. In Part 1, the exponential relationship between collector current and base-emitter voltage was verified, and the extracted thermal voltage and saturation current values were found to be consistent with expected theoretical ranges. This confirms the validity of the diode-based model for describing BJT operation in the active region.

In Part 2, the design and implementation of a common-emitter amplifier demonstrated how theoretical calculations can be translated into a working circuit. Although some discrepancies were observed between calculated and measured values, the transistor was successfully biased in the active region, validating the design approach. The frequency response analysis further

confirmed that the amplifier exhibits band-pass characteristics, with gain and bandwidth behaving as predicted by theory.

The addition of coupling capacitors and the emitter bypass capacitor clearly showed the trade-offs between gain, stability, and frequency response. In particular, the bypass capacitor significantly increased the gain, confirming the role of emitter degeneration in limiting amplification. These observations highlight that while theoretical models provide accurate predictions, real circuits are influenced by non-ideal effects such as component tolerances, transistor variability, and parasitic elements.

The experimental results validate the underlying theoretical assumptions and demonstrate their practical relevance. The experiment provides a clear understanding of how BJTs operate and how amplifier performance can be controlled through circuit design. These findings are essential for designing reliable analog circuits and emphasize the importance of considering both ideal and real-world factors in engineering applications.